Realization of High Break...
PDF

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Derive , simulate and plot Fig4 in detail

Thought for 17s >

Fig. 4 model

Figure 4 plots the theoretical avalanche breakdown voltage versus SOI silicon thickness t_s, for buried-oxide thicknesses $t_{ox} = 2,\ 4,\ 6\ \mu\mathrm{m}$, in the limit of very large lateral depletion width W. The source figure is on page 5, while the field and ionization-integral derivation is developed on pages 2–3. Realization of High Breakdown V...

1. Optimized electric field

The optimized RESURF condition requires the lateral electric field to be independent of lateral position:

$$\frac{\partial E_x}{\partial x} = 0.$$

The paper obtains the vertical field inside the depleted SOI layer as

$$E_y(x,y) = \frac{V x y}{t_{\mathrm{eff}} W t_s},$$

where

$$t_{\mathrm{eff}} = \frac{t_s}{2} + \frac{\varepsilon_{\mathrm{Si}}}{\varepsilon_{\mathrm{ox}}} t_{ox} \approx \frac{t_s}{2} + 3t_{ox}.$$

At the highest-field corner, $x = W,\ y = t_s$,

$$E_{\mathrm{Si,max}} = E_y(W, t_s) = \frac{V}{t_{\mathrm{eff}}}.$$

Thus,

$$V = E_{\mathrm{Si,max}} \left(\frac{t_s}{2} + 3t_{ox} \right).$$

This simple expression alone would predict that breakdown voltage increases with t_s. The upturn at very small t_s comes from the thickness dependence of the avalanche critical field.

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2. Avalanche ionization condition

The effective silicon ionization rate used by the authors is

$$\alpha(E) = a \exp\left(-\frac{b}{E} \right),$$

with

$$a = 7.03 \times 10^5 \ \mathrm{cm}^{-1}, \qquad b = 1.468 \times 10^6 \ \mathrm{V\ cm}^{-1}.$$

For $W \to \infty$, the horizontal ionization paths become negligible. Breakdown is determined by the vertical path at $x = W$.

Along this path,

$$E_y(W, y) = \frac{V y}{t_{\mathrm{eff}} t_s}.$$

The ionization integral is therefore

$$I_v = \int_0^{t_s} a \exp\left[-\frac{b\, t_{\mathrm{eff}} t_s}{V y} \right] dy.$$

Define

$$\gamma = \frac{b\, t_{\mathrm{eff}}}{V} = \frac{b}{E_{\mathrm{Si,max}}}.$$

Then

$$I_v = a \int_0^{t_s} \exp\left(-\frac{\gamma t_s}{y} \right) dy.$$

Using $u = \gamma t_s / y$,

$$I_v = a\gamma t_s \int_\gamma^\infty \frac{e^{-u}}{u^2}\, du.$$

Since

$$\int_\gamma^\infty \frac{e^{-u}}{u^2}\, du = \frac{e^{-\gamma}}{\gamma} - E_1(\gamma), \qquad \text{导出}$$

where

$$E_1(\gamma) = \int_\gamma^\infty \frac{e^{-u}}{u}\, du,$$

the vertical ionization integral becomes

$$I_v = a t_s \left[e^{-\gamma} - \gamma E_1(\gamma)\right].$$

Avalanche breakdown occurs when

$$I_v = 1.$$

Thus, for each silicon thickness t_s, solve

$$a t_s \left[e^{-\gamma} - \gamma E_1(\gamma)\right] = 1$$

numerically for γ. The corresponding breakdown voltage is

$$V_{\mathrm{BR}}(t_s, t_{ox}) = \frac{b}{\gamma(t_s)} \left(\frac{t_s}{2} + 3t_{ox}\right).$$

All lengths must be expressed in centimetres when using the quoted values of a and b.

3. Why the curves are U-shaped

There are two competing effects.

For large t_s, the effective voltage-supporting thickness

$$t_{\mathrm{eff}} = \frac{t_s}{2} + 3t_{ox}$$

increases with silicon thickness. The critical field changes relatively slowly, so

$$V_{\mathrm{BR}} \sim E_c\, t_{\mathrm{eff}}$$

increases with t_s.

For very small t_s, the avalanche path becomes extremely short. To make the ionization integral equal unity, the electric field must become much larger. Equivalently, the numerical solution gives a smaller $\gamma = b/E_c$, and therefore

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$$E_c = \frac{b}{\gamma}$$

rises sharply. Because the buried oxide converts that increased silicon critical field into voltage support over approximately $3t_{ox}$, the breakdown voltage rises as t_s decreases.

This produces the minimum in each curve.

Reproduced plot

Download the reproduced Fig. 4 plot

Download the reusable Python simulation

fig4_reproduction_data.csv

	A	B	C	D	E	F	G	H
1	t_ox (um)	minimum V	t_s at minir	V_BR at t_s	V_BR at t_s	V_BR at t_s=10 um (V)		
2	2.0	299.55124	3.4382665	862.95885	345.14232	344.1481184470764		
3	4.0	517.59878	6.0472808	1718.7858	663.73523	531.8652739636635		
4	6.0	717.49374	8.4559006	2574.6127	982.32814	719.5824294802507		
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Sheet1

Representative results from the reconstructed model are:

t_{ox}	Approximate minimum V_{BR}	t_s at minimum	V_{BR} at $t_s = 0.1\ \mu\mathrm{m}$
$2\ \mu\mathrm{m}$	$300\ \mathrm{V}$	$3.5\ \mu\mathrm{m}$	$863\ \mathrm{V}$
$4\ \mu\mathrm{m}$	$518\ \mathrm{V}$	$5.8\ \mu\mathrm{m}$	$1719\ \mathrm{V}$
$6\ \mu\mathrm{m}$	$717\ \mathrm{V}$	$8.5\ \mu\mathrm{m}$	$2575\ \mathrm{V}$

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The $t_{ox} = 2\ \mu\mathrm{m}$, $t_s = 0.1\ \mu\mathrm{m}$ prediction is somewhat above the paper's qualitative "about 700 V" statement, but it is consistent with the plotted curve's approximate graphical resolution and the simplifications used in the published analytical model.

Buried-oxide breakdown line

Continuity of normal electric displacement gives

$$E_{\mathrm{ox}} = \frac{\varepsilon_{\mathrm{Si}}}{\varepsilon_{\mathrm{ox}}} E_{\mathrm{Si,max}} \approx 3E_{\mathrm{Si,max}}.$$

Using

$$E_{\mathrm{Si,max}} = \frac{V}{t_s/2 + 3t_{ox}},$$

we obtain

$$E_{\mathrm{ox}} = \frac{3V}{t_s/2 + 3t_{ox}} = \frac{V}{t_{ox} + t_s/6}.$$

For the paper's assumed oxide critical field,

$$E_{\mathrm{ox,crit}} \approx 600\ \mathrm{V}/\mu\mathrm{m},$$

the dielectric limit is

$$V_{\mathrm{ox,BR}} = 600\left(t_{ox} + \frac{t_s}{6}\right)$$

with dimensions in micrometres. For $t_{ox} = 2\ \mu\mathrm{m}$ and thin SOI,

$$V_{\mathrm{ox,BR}} \approx 1200\ \mathrm{V},$$

which is the nearly horizontal dashed line in Fig. 4.

Sources

redo all for 4h-sic and GaN

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Thought for 18s

4H-SiC and GaN translation of Fig. 4

I reapplied the paper's thin-film RESURF derivation to idealized **4H-SiC-on-SiO_2** and **wurtzite GaN-on-SiO_2** structures. The geometry, uniform lateral-field assumption, and $W \to \infty$ limit are unchanged from the original silicon analysis.

Realization of High Breakdown V...

Download the combined 4H-SiC/GaN plot

Download the simulation script

fig4_4HSiC_GaN_data.csv

	A	B	C	D	E	F	G	H
1	material	tox_um	minimum_'ts_at_mini		VBR_at_0.(	VBR_at_0.1	VBR_at_1u	VBR_at_10um_k
2	4H-SiC	2.0	1.5578911(	2.0832484(	7.7960448	2.6074112	1.6202343(	2.02720894843(
3	4H-SiC	4.0	2.7811956	3.7296993(	15.584309	5.1890327	3.0933119(	3.04133295338(
4	4H-SiC	6.0	3.9205296	5.2536095	23.372574	7.7706542	4.5663894	4.05545695834
5	GaN	2.0	1.5300302(	1.8164668	6.4217005	2.4266827	1.5716526(	2.07538790040
6	GaN	4.0	2.7391659	3.3082584	12.836373	4.8270691	2.9881295	3.06578748826
7	GaN	6.0	3.8669370	4.6599730	19.251046	7.2274554	4.4046063	4.05618707612
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Sheet1

1. Generalized electrostatic derivation

Let the semiconductor relative permittivity be ε_s, and let the buried dielectric be SiO_2 with

$$\varepsilon_{ox} = 3.9.$$

Define

$$r_\varepsilon = \frac{\varepsilon_s}{\varepsilon_{ox}}.$$

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The optimized field distribution derived in the silicon paper remains

$$E_y(x,y) = \frac{Vxy}{Wt_s\left(t_s/2 + r_\varepsilon t_{ox}\right)}.$$

At the high-field corner $x = W,\ y = t_s$,

$$E_{s,\max} = \frac{V}{t_s/2 + r_\varepsilon t_{ox}}.$$

Therefore,

$$V = E_{s,\max}\left(\frac{t_s}{2} + r_\varepsilon t_{ox}\right)$$

and the effective voltage-supporting thickness is

$$t_{\text{eff}} = \frac{t_s}{2} + \frac{\varepsilon_s}{\varepsilon_{ox}}\, t_{ox}.$$

For the two materials used here:

$$r_\varepsilon(4H\text{-SiC}) = \frac{9.76}{3.9} \approx 2.503,$$

$$r_\varepsilon(\text{GaN}) = \frac{8.9}{3.9} \approx 2.282.$$

The 4H-SiC permittivity used is the reported ordinary relative permittivity 9.76, while the GaN calculation uses a static relative permittivity of 8.9. NSF Public Acce... +1

2. Avalanche model

As in the original paper, use a one-term Chynoweth ionization law:

$$\alpha(E) = a\exp\left(-\frac{b}{E}\right).$$

In the $W \to \infty$ limit, avalanche is governed by the vertical path at $x = W$:

$$I_v = \int_0^{t_s} a \exp\left[-\frac{b\, t_{\text{eff}} t_s}{V y}\right] dy.$$

Define

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$$\gamma = \frac{b\, t_{\text{eff}}}{V} = \frac{b}{E_{s,\max}}.$$

The integral reduces to

$$I_v = a t_s \left[e^{-\gamma} - \gamma E_1(\gamma)\right]$$

where

$$E_1(\gamma) = \int_\gamma^\infty \frac{e^{-u}}{u}\, du.$$

Breakdown occurs when

$$I_v = 1,$$

so γ is obtained from

$$a t_s \left[e^{-\gamma} - \gamma E_1(\gamma)\right] = 1.$$

The avalanche breakdown voltage is then

$$V_{\text{BR}} = \frac{b}{\gamma(t_s)} \left(\frac{t_s}{2} + \frac{\varepsilon_s}{\varepsilon_{ox}} t_{ox}\right).$$

Lengths must be converted to centimetres because a is in cm^{-1} and b is in V/cm.

3. Material parameters used

4H-SiC

The simulation uses the 300 K hole-impact-ionization fit

$$\alpha_p(E) = 5.31 \times 10^6 \exp\left(-\frac{1.31 \times 10^7}{E}\right)\ \text{cm}^{-1}.$$

Thus,

$$a_{\mathrm{SiC}} = 5.31 \times 10^{6}\ \mathrm{cm}^{-1},$$

$$b_{\mathrm{SiC}} = 1.31 \times 10^{7}\ \mathrm{V/cm}.$$

This fit was selected because holes have the smaller exponential field parameter and therefore dominate the simplified single-coefficient avalanche estimate over the relevant range. However, real 4H-SiC impact ionization is anisotropic and depends strongly on crystal direction. arXiv +1

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GaN

For c-axis GaN, the experimentally extracted hole coefficient is

$$\beta(E) = 7.13 \times 10^{6} \exp\left(-\frac{1.46 \times 10^{7}}{E}\right)\ \mathrm{cm}^{-1}.$$

Thus,

$$a_{\mathrm{GaN}} = 7.13 \times 10^{6}\ \mathrm{cm}^{-1},$$

$$b_{\mathrm{GaN}} = 1.46 \times 10^{7}\ \mathrm{V/cm}.$$

The measurements show that holes dominate at lower fields, while electron and hole coefficients become more comparable above roughly $3.3\ \mathrm{MV/cm}$. ResearchGate

4. Buried-SiO_2 limit

Normal displacement continuity gives

$$\varepsilon_s E_s = \varepsilon_{ox} E_{ox},$$

so

$$E_{ox} = r_\varepsilon E_s.$$

Combining this with the voltage expression,

$$V = E_{ox}\left(t_{ox} + \frac{t_s}{2r_\varepsilon}\right).$$

For an assumed SiO_2 dielectric strength of

$$E_{ox,\mathrm{crit}} = 600\ \mathrm{V}/\mu\mathrm{m},$$

the dielectric limit is

$$V_{ox,\mathrm{BR}} = 600 \left(t_{ox} + \frac{t_s}{2r_\varepsilon} \right).$$

For $t_{ox} = 2\ \mu\mathrm{m}$ and a very thin semiconductor film, both materials therefore approach approximately 导出

$$V_{ox,\mathrm{BR}} \approx 1.2\ \mathrm{kV}.$$

This dashed line is far below much of the intrinsic avalanche prediction. Consequently, a $2\ \mu\mathrm{m}$ SiO_2 layer prevents realization of most of the multi-kilovolt material capability.

5. Main simulation results

Material	t_{ox}	Minimum avalanche V_{BR}	Approximate t_s at minimum
4H-SiC	$2\ \mu\mathrm{m}$	about $1.5\ \mathrm{kV}$	a few $\mu\mathrm{m}$
4H-SiC	$4\ \mu\mathrm{m}$	about $2.8\ \mathrm{kV}$	a few $\mu\mathrm{m}$
4H-SiC	$6\ \mu\mathrm{m}$	about $3.9\ \mathrm{kV}$	several $\mu\mathrm{m}$
GaN	$2\ \mu\mathrm{m}$	about $1.4\ \mathrm{kV}$	a few $\mu\mathrm{m}$
GaN	$4\ \mu\mathrm{m}$	about $2.7\ \mathrm{kV}$	a few $\mu\mathrm{m}$
GaN	$6\ \mu\mathrm{m}$	about $3.8\ \mathrm{kV}$	several $\mu\mathrm{m}$

Representative calculated avalanche voltages are:

Material	t_{ox}	$t_s = 0.1\ \mu\mathrm{m}$	$t_s = 1\ \mu\mathrm{m}$	$t_s = 10\ \mu\mathrm{m}$
4H-SiC	$2\ \mu\mathrm{m}$	several kV	$1.62\ \mathrm{kV}$	$2.03\ \mathrm{kV}$
4H-SiC	$4\ \mu\mathrm{m}$	higher than $5\ \mathrm{kV}$	$3.09\ \mathrm{kV}$	$3.04\ \mathrm{kV}$
4H-SiC	$6\ \mu\mathrm{m}$	higher still	$4.57\ \mathrm{kV}$	$4.06\ \mathrm{kV}$
GaN	$2\ \mu\mathrm{m}$	several kV	$1.57\ \mathrm{kV}$	$2.08\ \mathrm{kV}$
GaN	$4\ \mu\mathrm{m}$	higher than $5\ \mathrm{kV}$	$2.99\ \mathrm{kV}$	$3.07\ \mathrm{kV}$

Material	t_{ox}	$t_s = 0.1\ \mu\mathrm{m}$	$t_s = 1\ \mu\mathrm{m}$	$t_s = 10\ \mu\mathrm{m}$
GaN	$6\ \mu\mathrm{m}$	higher still	$4.40\ \mathrm{kV}$	$4.06\ \mathrm{kV}$

6. Interpretation

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The U-shaped behavior survives for both wide-bandgap materials:

$$V_{\mathrm{BR}} = E_c(t_s)\left(\frac{t_s}{2} + r_\varepsilon t_{ox}\right).$$

At large t_s, the geometric voltage-supporting thickness grows, causing V_{BR} to rise.

At very small t_s, the avalanche path becomes so short that the required critical field rises sharply. That increasing E_c produces the left-hand upturn.

Compared with silicon, the entire curve shifts into the kilovolt regime because the exponential field parameters b for 4H-SiC and GaN are approximately an order of magnitude larger.

The practical result is that **the buried dielectric, not intrinsic semiconductor avalanche, becomes the dominant limitation** for much of the thin-film range. A realistic high-voltage implementation would therefore require a thicker dielectric, a higher-strength dielectric, a field-sharing substrate architecture, or some combination of those approaches.

These curves are analytical material translations rather than calibrated device predictions. They omit electron–hole coupled multiplication, nonlocal ionization, crystallographic anisotropy, polarization charge in GaN, interface charge, traps, defects, edge termination, and field-dependent dielectric reliability.

Sources

overlap Si , SiC and GaN in 1 graph

Thought for 27s

Done — I overlaid **Si, 4H-SiC, and GaN** on a **single Fig. 4-style graph**.

Files

- Overlapped comparison plot 导出
- fig4_overlap_Si_SiC_GaN_data.csv

	A	B	C	D	E	F	G	H
1	material	tox_um	min_VBR_k	ts_at_min_	VBR_at_0.1	VBR_at_1u	VBR_at_10um_kV	
2	Si	2.0	0.2995511	3.4235590	0.8629588	0.3451423	0.3441481184470764	
3	4H-SiC	2.0	1.5578911	2.0832484	2.6074112	1.6202343	2.0272089484307645	
4	GaN	2.0	1.5300302	1.8164668	2.4266827	1.5716526	2.075387900402876	
5	Si	4.0	0.5175972	6.0251988	1.7187858	0.6637352	0.5318652739636636	
6	4H-SiC	4.0	2.7811956	3.7296993	5.1890327	3.0933119	3.0413329533863904	
7	GaN	4.0	2.7391659	3.3082584	4.8270691	2.9881295	3.065787488262693	
8	Si	6.0	0.7174927	8.3428832	2.5746127	0.9823281	0.7195824294802506	
9	4H-SiC	6.0	3.9205296	5.2536095	7.7706542	4.5663894	4.055456958342017	
10	GaN	6.0	3.8669370	4.6599730	7.2274554	4.4046063	4.0561870761225105	
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Sheet1

- Python script

Notes

- The graph includes **all three materials** for $t_{ox} = 2,\ 4,\ 6\ \mu m$ on the same axes.
- Line style distinguishes material:
  - **Solid** = Si
  - **Dashed** = 4H-SiC
  - **Dash-dot** = GaN
- The y-axis is in **kV** so the Si, SiC, and GaN curves can be compared directly.

Quick takeaway

- **Si** stays in the **sub-kV to ~1 kV** range here.
- **4H-SiC** and **GaN** shift upward into the **multi-kV** regime.
- For the same t_{ox}, **4H-SiC and GaN are close**, with **4H-SiC slightly higher** in this idealized model.

If you want, I can also make a **cleaner version with only $t_{ox} = 2\ \mu m$** for all three materials, which is easier to read visually.

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py code?

Download the Python code

why Si has lrager plateau at the bottom?

Thought for a few seconds

Si shows a broader, flatter minimum because in this model its **critical avalanche field changes more gradually with thickness** than it does for 4H-SiC or GaN.

The breakdown voltage is

$$V_{\mathrm{BR}}(t_s) = E_c(t_s)\left(\frac{t_s}{2} + \frac{\varepsilon_s}{\varepsilon_{ox}}\,t_{ox}\right),$$

where:

- the geometric factor increases with t_s,
- the critical field $E_c(t_s)$ decreases as the avalanche path becomes longer.

The bottom of the U-curve occurs where these two trends nearly cancel.

Why the cancellation lasts longer for Si

The ionization law is

$$\alpha(E) = ae^{-b/E}.$$

The parameters used were approximately

$$b_{\mathrm{Si}} = 1.47 \times 10^6\ \mathrm{V/cm},$$

$$b_{\mathrm{SiC}} = 1.31 \times 10^7 \ \mathrm{V/cm}, \qquad b_{\mathrm{GaN}} = 1.46 \times 10^7 \ \mathrm{V/cm}.$$

The b parameter for SiC and GaN is almost ten times larger than for Si. Consequently, their ionization rates are much more sensitive to a change in electric field. A modest change in t_s therefore causes a relatively strong change in the required critical field.

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For Si, the required field varies more gently with t_s. Over a relatively wide thickness range,

$$E_c(t_s) \downarrow$$

almost compensates

$$\left(\frac{t_s}{2} + \frac{\varepsilon_s}{\varepsilon_{ox}} t_{ox} \right) \uparrow .$$

Therefore V_{BR} changes only slightly, producing the broad plateau.

Logarithmic-slope interpretation

At the minimum,

$$\frac{dV_{\mathrm{BR}}}{dt_s} = 0.$$

Equivalently,

$$\frac{d \ln E_c}{d \ln t_s} = - \frac{t_s/2}{t_s/2 + (\varepsilon_s/\varepsilon_{ox}) t_{ox}}.$$

A broad plateau means that the two logarithmic slopes remain close over a wide interval.

For Si, the slope of $E_c(t_s)$ changes gradually, so this condition remains approximately satisfied for many thickness values.

For SiC and GaN, $E_c(t_s)$ changes more sharply, so the cancellation occurs over a narrower range and the U-shaped minimum is more pronounced.

Permittivity is a secondary effect

Si also has a larger dielectric ratio:

$$\frac{\varepsilon_{\mathrm{Si}}}{\varepsilon_{\mathrm{ox}}} \approx 3,$$

compared with approximately

$$2.50 \quad \text{for 4H-SiC}, \qquad 2.28 \quad \text{for GaN}.$$

This makes the oxide-related term somewhat more dominant for Si and further weakens the relative influence of $t_s/2$, contributing slightly to the flatter bottom.

The main reason, however, is the much weaker field sensitivity of the Si impact-ionization coefficient. The original paper's U-shaped silicon behavior follows this same competition between geometrical voltage support and thickness-dependent critical field.

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Realization of High Breakdown V...

Sources

$$\frac{d\ln E_c}{d\ln t_s} = -\frac{t_s/2}{t_s/2 + (\varepsilon_s/\varepsilon_{ox})t_{ox}}.$$

plot left hand side vs ts with material and tox

Thought for 5s

Plotted the logarithmic critical-field slope together with the thickness/oxide term for **Si, 4H-SiC, and GaN**, using $t_{ox} = 2,\ 4,\ 6\ \mu\mathrm{m}$.

Download the plot

log_slope_intersections_material_tox.csv

	A	B	C	D	E	F	G	H
1	material	tox_um	intersectio	lhs_at_inte	rhs_at_intersection			
2	Si	2.0	3.4190510	-0.222437	-0.22174198675831863			
3	Si	4.0	6.0291697	-0.200189	-0.2007771041147908			
4	Si	6.0	8.4035641	-0.189050	-0.18925427057784822			
5	4H-SiC	2.0	2.0777656	-0.171040	-0.17188632324980072			
6	4H-SiC	4.0	3.7149851	-0.157027	-0.15651599771469063			
7	4H-SiC	6.0	5.2501495	-0.149693	-0.14880988719326466			
8	GaN	2.0	1.8345067	-0.166692	-0.1673405967159904			
9	GaN	4.0	3.2800451	-0.153333	-0.15230199815287246			
10	GaN	6.0	4.7000556	-0.146053	-0.1464891344315043			
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Sheet1

导出

The blue line is

$$\frac{d\ln E_c}{d\ln t_s},$$

and the dashed lines are

$$-\frac{t_s/2}{t_s/2+(\varepsilon_s/\varepsilon_{ox})t_{ox}}.$$

The marked intersections satisfy

$$\frac{d\ln E_c}{d\ln t_s}=-\frac{t_s/2}{t_s/2+(\varepsilon_s/\varepsilon_{ox})t_{ox}},$$

so they correspond approximately to the minimum of each breakdown-voltage curve.

One important point: the blue left-hand-side curve depends on **material and** t_s, but not directly on t_{ox}. The oxide thickness only moves the dashed right-hand-side curve and therefore changes the intersection thickness.